Average (Measure of Central Tendency) Chapter II

Average: An average is a single value within the range of data which is used to represent all values of the series. Since an average value lies somewhere in the centre of data, it is also called as measure of central tendency.

Objectives of average:

1. To get a single value which represent the given set of data.
2. To facilitate the comparison between two or more than two series.

Properties of good average:

- ✓ It should be simple to understand.
- ✓ It should be easy to compute.
- ✓ It should be based on all the observations.
- ✓ It should not be affected by extreme value.
- ✓ It should be properly defined.
- ✓ It should be capable for further mathematical treatment.
- ✓ It should have sampling stability.

Types of average:

1. Arithmetic mean.
2. Median.
3. Mode.
4. Geometric mean.
5. Harmonic mean.

1. Arithmetic mean (or mean): Mean of a set of observations is sum of observations divided by total number of observations.

There are three types of series

a. Individual series or ungrouped data

b. Discrete frequency distribution

c. Continuous frequency distribution

a. Individual series: Suppose X_1, X_2, X_3…….. Xn be the n observations, mean is the sum of n observations divided by total number of observations.

i) Direct method:

Mean ($\bar{X}$) = $\frac{(X_1+X_2+X_3+\cdots\ldots\ldots\ldots+X_n)}{n} = \frac{\sum_{i=1}^{n} X_i}{n}$

Example 2.1: Height of 5 trees are 62, 68, 66, 71 and 73 inches. Find mean height of the tree.

Solution:

$\bar{X} = \frac{62+68+66+71+73}{5} = \frac{340}{5} = 68$ inches.

ii) **Shortcut method:** Either by change of origin or change of scale. In shortcut method when data value is large, calculations is simplified either changing the origin or scale or both.

Change of origin: Either add or subtract a common value (Assumed value) from each observation, which lies somewhere in the middle of given set of data.

$\bar{X} = A + \frac{\sum d}{n}$, where d = X - A

Find mean of Example 2.1 by changing the origin

X	**d = X – A (A = 67)**
62	-5
68	1
66	-1
71	4
73	6
Total	5

$= 67 + \frac{5}{5} = 68$ inches

Change of scale: Either multiply or divide each observation by a common value (h).

If each observation is divided by h

$\bar{X} = h\frac{\sum d}{n}$, where $d = \frac{X}{h}$

If each observation is multiplied by h

$\bar{\mathbf{X}} = \frac{\sum d}{hn}$, where d = hX

Change of origin and change of scale: Subtract a common value (A) from each observation and divide this deviation from a common value (h)

$\bar{\mathbf{X}} = A + h\frac{\sum d'}{n}$, where $d' = \frac{X-A}{h}$

Example 2.2: Find mean for following set of data

X : 5 15 25 35 45 55 65

Solution:

Observation number	**X**	**change of scale** $\mathbf{d=\frac{X}{h}}$ **(h =5)**	**Change of Origin** **d = (X – A), A =45**	**Change of origin and change of scale** $\mathbf{d=\frac{X-A}{h}}$ **(A=45, h =10)**
1	5	1	-40	- 4
2	15	3	-30	-3
3	25	5	-20	-2
4	35	7	-10	-1
5	45	9	0	0
6	55	11	10	1
7	65	13	20	2
Total	245	49	-70	-7

Direct Method Mean $\bar{\mathbf{X}} = \frac{245}{7} = 35$

Change of Origin Mean $\bar{\mathbf{X}} = 45 + \frac{-70}{7} = 45 - 10 = 35$

Change of Scale Mean $\bar{\mathbf{X}} = 5 \times \frac{49}{7} = 5 \times 7 = 35$

Change of Origin and change of scale Mean $\bar{\mathbf{X}} = 45 + 10\frac{(-7)}{7} = 45 - 10 = 35$

Mathematical properties of Mean:

i) Sum of deviation of observations from the mean is zero i.e. $\sum(X - \bar{X}) = 0$.

ii) Sum of square of deviation of observation from mean is minimum as compared to any other value i.e. $\sum(X - \bar{\mathbf{X}})^2$.

iii) If there are two or more than two series and number of observation and mean for these series is given, we can find the combined mean for both the series. If n_1 and n_2 are number of observations in first and second series respectively and $\bar{X}_1$ and $\bar{X}_2$ are mean of these series respectively, combined mean of both the series is

$$\bar{\mathbf{X}} = \frac{\mathbf{n_1\bar{X}_1 + n_2\bar{X}_2}}{\mathbf{n_1 + n_2}}$$

Verification of mathematical properties:

X	$\mathbf{(X - \bar{X}),\ \bar{X} = 35}$	$\mathbf{(X - \bar{X})^2}$	**(X − A), A = 45**	$\mathbf{(X - A)^2}$
5	-30	900	-40	1600
15	-20	400	-30	900
25	-10	100	-20	400
35	0	0	-10	100
45	10	100	0	0
55	20	400	10	100
65	30	900	20	400
Sum	0	2800		3500

Mean = $\frac{245}{7} = 35$

i) Sum of deviation of observations from their mean is always equal to 0.

ii) Sum of square of deviation of observations from mean is 2800 and sum of square of deviation of observation from 45 (other than mean) is 3500

Example 2.3: Mean height of 30 plants of variety A is 60 inches, whereas mean height of 20 plants of variety B is 45 inches. Find combined mean height of both the varieties.

Solution $\bar{\mathbf{X}} = \frac{30 \times 60 + 20 \times 45}{30+20} = \frac{2700}{50} = 54$ inches

b) Discrete frequency distribution: Suppose X is a variable taking values X_1, X_2, X_3,……., X_n with corresponding frequencies as f_1, f_2,f_3,……, f_n.

X	X_1	X_2	X_3	.	.	.	.	.	.	X_n	Total
f	f_1	f_2	f_3	.	.	.	.	.	.	f_n	N
fX	f_1x_1	f_2x_2	f_3x_3	.	.	.	.	.	.	f_nX_n	$\sum \mathbf{fx}$

Direct Method: $\bar{\mathbf{X}} = \frac{f_1X_1+f_2X_2+f_3X_3\ldots\ldots\ldots+f_nX_n}{f_1+f_2+f_3+\cdots\ldots\ldots+f_n} = \frac{\sum \mathbf{fx}}{\mathbf{N}}$

Shortcut method:

i) By changing origin

$$\bar{\mathbf{X}} = \mathbf{A} + \frac{\sum \mathbf{fd}}{\mathbf{N}}, \text{ where } \mathbf{d = X - A}$$

ii) Change of origin and change of scale:

$$\bar{\mathbf{X}} = \mathbf{A} + \mathbf{h}\frac{\sum \mathbf{fd'}}{N}$$

Example 2.4 : 20 sets of seeds were kept for germination and in each set there were 10 seeds, number of seeds germinated in each set were recorded and following is the frequency distribution of number of seeds germinated. Find mean number of seeds germinated.

Number of seeds germinated (X)	**f**	**fX**	**d=X-A, A =7**	**fd**
5	2	10	-2	-4
6	5	30	-1	-5
7	6	42	0	0
8	4	32	1	4
9	2	18	2	4
10	1	10	3	3
Total	20	142		2

Solution:

i) Direct Method: Mean $\bar{\mathbf{X}} = \frac{142}{20} = 7.1$

ii) Change of origin : Mean $\bar{\mathbf{X}} = 7 + \frac{2}{20} = 7 + 0.1 = 7.1$

c) Continuous frequency distribution: Find mid value of each class and rest of the procedure is same as in discrete frequency distribution.

Example 2.5 : Find mean for following continuous frequency distribution

Class	f	m.v.(X)	Fx	d=X-A	fd	$d' = \frac{X-A}{h}$	fd′
0-10	3	5	15	-20	-60	-2	-6
10-20	5	15	75	-10	-50	-1	-5
20-30	7	25	175	0	0	0	0
30-40	4	35	140	10	40	1	4
40-50	1	45	45	20	20	2	2
Total			450		50		-5

Solution:

1. **Direct Method** :

   $\bar{X} = \frac{\sum fx}{N} = \frac{450}{20} = 22.5$

2. **By changing origin:**

   $\bar{X} = A + \frac{\sum fd}{N} = 25 + \frac{(-50)}{20} = 22.5$

3. **By changing origin and scale**:

   $\bar{X} = A + h\frac{\sum fd'}{N} = 25 + \frac{10(-5)}{20} = 22.5$

Merits of mean:

- ✓ It is simple to understand and easy to calculate.
- ✓ It is based on each and every observation.
- ✓ It is capable for further mathematical treatments i.e. we can find combined mean.
- ✓ Properly defined.
- ✓ It is least affected by sampling fluctuations.

Limitations:

- ✓ Extreme value affects the value of mean.
- ✓ In case of open end distribution, we cannot compute the value of mean, however by making assumption we can calculate mean.e.g. If in a continuous frequency distribution, last class is greater than 90 and class interval is 10, it is assumed that last class is 90-100.
- ✓ It can not be used for qualitative data.

2. Median: Median is that value which divide the whole set of data in two equal parts.

a. Individual series: There are two cases.

i. **If number of observation is odd**: Arrange the data either in increasing or decreasing order.

Median = $\frac{\mathbf{n+1}}{\mathbf{2}}$ **th** term.

Example 2.6 : Find median for given set of data 5, 9, 8, 4, 7, 6, 3

Solution : Arrange given set of data in increasing order i.e. 3,4,5,6,7,8,9.

Number of observations =7

$\frac{7+1}{2} = 4^{th}$ term is 6.

Median = 6.

ii. **If the number of observation is even**: Arrange the given data either in decreasing or increasing order. In this case there are two middle terms and they are $\frac{\mathbf{n}}{\mathbf{2}}$ **th** and $\frac{\mathbf{n}}{\mathbf{2}} + \mathbf{1^{th}}$ term. Median is the mean of two middle terms.

Example 2.7: Find median for given set of data 5, 9, 8, 4, 7, 6, 3, 2

Solution: Arrange given set of data in increasing order i.e. 2, 3,4,5,6,7,8,9.

n = 8, two middle terms are $\frac{8}{2} = 4\text{th}$ and $\frac{8}{2} + 1 = 5\text{th}$

4th terms is 5 and 5th terms is 6.

Median = $\frac{5+6}{2} = 5.5$

b. Discrete frequency distribution:

i) Find cumulative frequency,

ii) Find $\frac{N}{2}$,

iii) The variable whose cumulative frequency is just greater than $\frac{N}{2}$,will be the value of median

Example2.8: Find median for following discrete frequency distribution.

X	f	C.F.
1	5	5
2	8	13

3	9	22
4	12	34
5	6	40
6	7	47
7	4	51
	N =51	

Solution:

$\frac{N}{2}$ = 25.5, C.F. just greater than 25.5 is 34 and corresponding value of the variable X is 4. Hence median is **4**.

c) Continuous frequency distribution:

i) Find cumulative frequency of each class.

ii) Find $\frac{N}{2}$.

iii) The class whose cumulative frequency is just greater than $\frac{N}{2}$ is median class.

iv) median is obtained as

$$\text{Median} = l + h\frac{(\frac{N}{2} - c.f)}{f}$$

where l = lower limit of median class

h = class interval

N = Total frequency

c.f. = cumulative frequency of class just preceding to median class

f = frequency of median class.

Example2.9: Find median of following continuous frequency distribution

Class	**f**	**C.F.**
0-10	3	3
10-20	6	9

20-30	12	21
30-40	7	28
40-50	2	30

Solution:

$\frac{N}{2} = \frac{30}{2} = 15$, C.F. just greater than 15 is 21 and corresponding class is 20-30.

Median class = 20-30, l = 20, h = 30 - 20 =10, f = 12 and C.F. = 9

Therefore, Median $= 20 + 10\frac{(\frac{30}{2} - 9)}{12} = 20 + 5 = 25$

Mathematical properties of Median:

1. Sum of deviation of observation from median ignoring sign is minimum as compared to any other value.

   Verification: In following set of data, n = 7. median is 6

X	**X - 5**	**X - 6 (median)**	**X-7**
3	-2	-3	-4
4	-1	-2	-3
5	0	-1	-2
6	1	0	-1
7	2	1	0
8	3	2	1
9	4	3	2
Total	13	12(minimum)	12

Merits of Median:

- ✓ Extreme values do not affect the value of median.
- ✓ In case of open end distribution, the value of median can be obtained.

- ✓ In case of qualitative data, the value of median can be obtained easily.
- ✓ Value of median can be obtained graphically.

Limitations:

- ✓ Require arranging of data.
- ✓ Value of median is not dependent on each and every observation. In some cases we can't find the value of median as in even number of observation, the median is the mean of two middle terms.

Other partition value

i) Quartile: which divide the given set of data in four equal parts.

ii) Decile: which divide the given set of data in ten equal parts.

iii) Percentile: which divide given set of data in hundred equal parts.

i) Quartile: Three quartiles are denoted by Q_1, Q_2 and Q_3.

For finding Q_1 , find $\frac{N}{4}$, class whose C.F. is just greater than $\frac{N}{4}$ will be the first quartile class. Q_1 is given by:

$$\mathbf{Q_1 = l + h\frac{\left(\frac{N}{4} - C.F.\right)}{f}}$$

For finding Q_2, find $\frac{2N}{4}$, class whose C.F. is just greater than $\frac{2N}{4}$ will be the second quartile class. Q_2 value is given by:

$$\mathbf{Q_2 = l + h\frac{\left(\frac{2N}{4} - C.F.\right)}{f}}$$

For finding Q_3, find $\frac{3N}{4}$, class whose C.F.is just greater than $\frac{3N}{4}$ will be the third quartile class. Q_3 value is given by:

$$\mathbf{Q_3 = l + h\frac{\left(\frac{3N}{4} - C.F.\right)}{f}}$$

ii) Decile: Nine points are denoted by D_1, D_2, D_3 …… , D_9.

$\mathbf{D_1 = \frac{N}{10}}$, the class whose C.F.is just greater than $\frac{N}{10}$ will be the first decile class. D_1 is obtained by:

$$\mathbf{D_1 = l + h\frac{\left(\frac{N}{10} - C.F.\right)}{f}}$$

Similarly $D_6 = \frac{6N}{10}$, then the class whose C.F. is just greater than $\frac{6N}{10}$ will be the sixth decile class. D_6 is obtained by:

$$\mathbf{D_6 = l + h\frac{(\frac{6N}{10} - C.F.)}{f}}$$

iii) Percentile: 99 points are denoted by P_1, P_2, P_3, ……., P_{99}.

$\mathbf{P_1 = \frac{N}{100}}$, then the class whose C.F. is just greater than $\frac{N}{100}$ will be the first percentile class. Therefore, P_1 is obtained by:

$$\mathbf{P_1 = l + h\frac{(\frac{N}{100} - C.F.)}{f}}$$

$$\mathbf{P_{33} = l + h\frac{(\frac{33N}{100} - c.f)}{f}}$$

3. Mode: Mode is that value which occurs most frequently in the distribution or which is having maximum frequency.

Discrete frequency distribution: Find variable whose frequency is maximum, that variable will be value of mode

Example 2.10: Find mode for following discrete frequency distribution

Solution :

X	f
1	6
2	8
3	7
4	5
5	2

Maximum frequency is 8 and corresponding value of the variable is 2. Hence mode of the distribution is 2.

Continuous frequency distribution: The class whose frequency is maximum is known as modal class. Mode is obtained by

$$\textbf{Mode} = \mathbf{l} + \mathbf{h}\frac{(\mathbf{f_1}-\mathbf{f_0})}{(\mathbf{2f_1}-\mathbf{f_0}-\mathbf{f_2})}$$

where, l = lower limit of modal class

h = class interval ,

f_1 = maximum frequency (frequency of modal class)

f_0 = frequency of class just preceding to modal class

f_2 = frequency of class just succeeding modal class

Example 2.11: Find mode of following distribution.

Class	F
0-10	3
10-20	6
20-30	12
30-40	7
40-50	2
N	30

Solution:

Maximum frequency = 12 (f_1)

Modal class = 20-30.

L =20; h = 30-20 =10; f_0 = 6; f_2 = 7

Mode = $20 + 10\frac{(12-6)}{(24-6-7)}$

$= 20 + 10\left(\frac{6}{11}\right) = 20 + 5.45 = 25.45$

Merits: Same as median

Limitations:

1. In some case, we cannot find single value of mode. Some distribution is having two modes. When there are two modes, such distribution is known as Bimodal distribution.

4. Geometric Mean:

Ungrouped data: Suppose $X_1, X_2, X_3 \ldots\ldots X_n$ be n observation, geometric mean is defined as n^{th} root of product of n observation.

$$\textbf{G.M.} = (\mathbf{X_1} . \mathbf{X_2} . \mathbf{X_3} \ldots\ldots\ldots\ldots . . \mathbf{X_n})^{\frac{1}{n}}$$

Example 2.13: Find Geometric mean of 2, 4, 8.

Solution:

$$G.M = (2 \times 4 \times 8)^{\frac{1}{3}} = 64^{\frac{1}{3}} = 4$$

When n is large, GM is calculated with the help of log:

$$\log G.M. = \frac{1}{n}(\log X_1 + \log X_2 + \log X_3 + \ldots\ldots \log X_n)$$

$$\log G.M. = \frac{1}{n}(\sum \log x)$$

$$\textbf{G.M. = antilog } (\frac{1}{n} \sum \textbf{logx})$$

G. M. is generally used when there is a percentage increase or decrease in the given set of data.

Example 2.14: Percentage increase in the price in the year 2010 is 20%, 2011 is 30% and in 2012 is 40%. What will be the mean percentage increase in price during three years?

Solution: Let the price taken in previous year is 100

Year	% increase in price	Price X	LogX
2010	20	120	2.07
2011	30	130	2.11
2012	40	140	2.14
Total			6.32

$$G.M. = \text{antilog } \frac{6.32}{3} = \text{antilog } (2.107) = (10)^{2.107}$$

G.M. = 127.84

Therefore, percentage increase in price

= 127.84– 100 = 27.84 %

Note: If one of the observations is zero, then we cannot find the value of G.M.

Merits:

- ✓ Based on each and every observation
- ✓ Properly defined
- ✓ Capable for further mathematical treatments

- ✓ Gives more weight to smaller observation and less weight to larger observations.

Demerits:

- ✓ It is not easy to understand and is difficult to compute.
- ✓ Cannot be calculated if any of the observation is zero or if there are both positive and negative values

5. **Harmonic Mean:** Harmonic Mean is the reciprocal of arithmetic mean of the reciprocal of the observations.

$$H.M = \frac{1}{\frac{1}{n}\left(\frac{1}{X_1}+\frac{1}{X_2}+\frac{1}{X_3}+ \ldots.+\frac{1}{X_n}\right)} = \frac{n}{\sum_{i=1}^{n}\frac{1}{X_i}}$$

Harmonic mean is generally used when rate is given. e.g. speed in km/hr.

Example 2.15: A person moves from one place to another place with average speed of 20 km/hr and return with average speed of 30 km/hr. What will be his average speed during whole journey?

Solution: n = 2

$$H.M. = \frac{2}{\frac{1}{30}+\frac{1}{20}} = \frac{2}{\frac{5}{60}} = 2 \times 12 = 24 \text{ km/hr}$$

Verification: Let the distance between two places A and B is 100 km

Time covered from A to B = $\frac{100}{20}$ = 5 hours

Time covered from B to A = $\frac{100}{30} = \frac{10}{3}$ hours

Total distance = 200 km

Time = $5 + \frac{10}{3} = \frac{25}{3}$

Speed = $\frac{200}{\frac{25}{3}}$ = 24 km/hour

Merit and demerits of Harmonic mean are same as that of Geometric mean**.**

Relationship between A.M., G.M. and H.M.:

A.M. ≥ G.M. ≥ H.M.

The sign of equality will hold true when all observations are same.

Relationship between Mean, median and mode

Mode = 3 Median – 2 Mean

Example 2.16: In a given distribution mean =35 and mode = 30, find median

Solution: 30 = 3×Median - 2×35

3 Median = 100

Median = 33.33

Example 2.17: Two numbers are 2 and 8. Find Arithmetic mean, geometric mean and harmonic mean

Solution:

AM $= \frac{2+8}{2} = 5$

GM $= (2 \times 8)^{\frac{1}{2}} = 4$

HM $= \frac{2}{\frac{1}{2}+\frac{1}{8}} = \frac{2}{\frac{5}{8}} = 3.2$

Example 2.18: Mean, median and mode of 100 observations was found to be 50, 52 and 53. Later on it was found that one of the observation which was 85 was misread as 58. Find correct value of mean, median and mode.

Solution:

Mean $= \frac{\Sigma X}{100}$

Uncorrected Total ΣX = 100X50 =5000

Corrected Total = 5000-58 +85 =5027

Corrected Mean $= \frac{5027}{100} = 50.27$

Since extreme values do not affect the value of median and mode, median and mode remains same. i.e. true value of median and mode are 52 and 53 respectively.

Measure of Dispersion, Skewness and Kurtosis Chapter III

3.1 Dispersion: It measures the variability or scatteredness of mass of figures in the given set of data from its average.

Let there be three series:

Observation number	Series I	Series II	Series III
1	10	8	1
2	10	9	5
3	10	10	10
4	10	11	15
5	10	12	19
Total	50	50	50
Mean	10	10	10
Range	0	4	18

Mean of all series = $\frac{50}{5}$ = 10

If it is given that mean of 5 observations is 10, we can not have any idea about the distribution because in 1st series all the observations are same, in 2nd series values lies between 8 to 12 and in 3rd series values lies between 1 to 19. If there is variability in given set of data, average value must be supported by some measure of dispersion. Average value gives idea about concentration of observations about central part of the distribution. If we know only average value, we can not have complete idea about the distribution. Thus measure of central tendency must be supported by some measure of dispersion.

Properties of good measure of Dispersion:

- Simple to understand
- Easy to compute
- Based on all observations
- Properly defined
- Capable for further mathematical treatments

- It must have sampling stability.

Types of dispersion: Measure of Dispersion is classified into two category

A) Absolute measure of dispersion
B) Relative measure of dispersion

A) Absolute measure of dispersion

1. Range.
2. Quartile deviation.
3. Mean deviation.
4. Standard deviation.

1. Range:

Individual Series: Range is the difference between maximum and minimum value.

In above example range of the 1st, 2nd and 3rd series is 0, 4 and 18 respectively

Continuous frequency distribution: There are two methods to find out the range:

Class interval	Mid value	frequency
0-10	5	5
10-20	15	8
20-30	25	13
30-40	35	24
40-50	45	46
50-60	55	52
60-70	65	34
70-80	75	19
80-90	85	8
90-100	95	2

1. Range is difference between the mid value of last class and mid value of the first class. Range is 90.
2. Range is difference between the upper limit of last class and the lower limit of first class. Here, range is 100.

Same method must be used to compare different series,.

Merits of range:

- ✓ Simple to understand, easy to calculate.
- ✓ Takes minimum time to compute

Limitations:

- ✓ Range involves only two extreme values.
- ✓ It does not tell about distribution of series within two extreme values.

2. **Quartile deviation**:

$$\mathbf{Q_D = \frac{Q_3 - Q_1}{2}}$$

Inter quartile range = $Q_3 - Q_1$

where Q_1 and Q_3 are first and third quartile

Merits:

- ✓ Simple to understand, easy to calculate
- ✓ In open end distribution quartile deviation can be obtained.

Limitations:

- ✓ It ignores 50% of the observation
- ✓ Not capable for further mathematical treatments.

3. **Mean deviation**: Mean deviation is the arithmetic mean of absolute value of deviation of observation from its average (Mean, median or mode),

   a. **Ungrouped data**: Suppose $X_1, X_2, X_3, \ldots X_n$ be n observations

M.D. $= \frac{1}{n} \sum |X - A|$ where A = Average value (mean, median or mode).

Note: Mean deviation is minimum when deviation is taken from median.

Merits:

- ✓ Simple to compute.
- ✓ Easy to understand.
- ✓ Less affected by extreme values as compared to range.
- ✓ In case of open end distribution, mean deviation from median or mode can be computed.
- ✓ Based on each and every observation.

Limitations:

- ✓ M.D. is minimum when taken from median but median is not a best measure of central tendency.
- ✓ Not capable for further mathematical treatments.
- ✓ Algebraic signs are ignored.

4. **Standard deviation (σ):** It is the positive square root of arithmetic mean of square of deviation of observation from the mean. It is denoted by σ .

   a) Individual series: Suppose $X_1, X_2, X_3, \ldots\ldots\ldots, X_n$ be n observations

$$\sigma = \sqrt{\frac{\sum(X - \overline{X})^2}{n}}$$

Example 3.2 : Find standard deviation for the number 5, 8,12,9 and 1

Solution:

Observation number	X	$(X - \overline{X})$	$(X - \overline{X})^2$	X^2
1	5	-2	4	1
2	8	1	1	4
3	9	2	4	9
4	12	5	25	16
5	1	-6	36	25
Total	35	0	70	315

$$\overline{X} = \frac{35}{5} = 7$$

$$\sigma = \sqrt{\frac{1}{5} \times 70} = \sqrt{14} = 3.74$$

When the mean is in fraction and the value of X is small, then σ is calculated as

$$\sigma = \sqrt{\frac{1}{n}\left(\sum X^2 - \frac{(\sum X)^2}{n}\right)}$$

$$\sigma = \sqrt{\frac{315 - \frac{35^2}{5}}{5}} = \sqrt{\frac{315 - 245}{5}} = \sqrt{14} = 3.74$$

When the mean is in fraction and the value of X is large:

Standard deviation is independent of change of origin.

Observation Number	X	d = X – A, A =8	d^2
1	5	-3	9
2	8	0	0
3	9	1	1
4	12	4	16
5	1	-7	49
Total	35	-5	75

$$\sigma = \sqrt{\frac{1}{n}\left[\sum d^2 - \frac{(\sum d)^2}{n}\right]} = \sqrt{\frac{1}{5}\left[75 - \frac{(-5)^2}{5}\right]} = \sqrt{14} = 3.74$$

Change of origin and change of scale: Standard deviation is independent of change of origin but not of scale

Merits:

- ✓ Based on each and every observation
- ✓ Capable for further mathematical treatments

Limitations:

- ✓ Difficult to compute.

Note : All measure of dispersion have same unit in which observation is measured.

Variance (σ^2): It is the arithmetic mean of square of deviation of observation from mean.

Suppose $X_1, X_2, X_3, \ldots\ldots\ldots\ldots X_n$ be n observation.

$$\sigma^2 = \frac{1}{n}\sum (X - \bar{X})^2$$

Note :

- Variance is independent of change of origin but not of change of scale
- Variance will be zero if all the observations are same

B) Relative measure of Dispersion

When two or more than two series to be compared, which differ in their units. Coefficient of Dispersion (COD) is calculated to make the series unit free.

1. Coefficient of Dispersion based on range:

$$\text{COD} = \frac{L - S}{L + S}$$

2. Coefficient of Dispersion based on quartile deviation:

$$\text{COD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

3. Coefficient of Dispersion based on mean deviation:

$$\text{COD} = \frac{\text{Mean Deviation}}{\text{Average from which mean deviation is calculated}}$$

4. Coefficient of Dispersion based on standard deviation:

$$\text{COD} = \frac{\sigma}{\bar{X}}$$

Coefficient of variation(CV): 100 times coefficient of dispersion based upon standard deviation is known as coefficient of variation. It measures the percentage variation in the mean, and standard deviation is considered as the total variation.

The series having smaller value of C.V. is more consistent or having smaller variability. The series having larger value of C.V. have more variability and less consistent.

$$\text{C.V} = \frac{\sigma}{\bar{X}} \times 100$$

Example 3.5: Following table gives the mean weight and variance of 100 female and 50 male. Which series is more consistent?

Parameter	Female	Male
N	100	50
Mean	45	60
Variance	9	4
Coefficient of Variation	$\sigma = 3$ $= \frac{3}{45} \times 100$ $= 6.6$	$\sigma = 4$ $= \frac{2}{60} \times 100$ $= 3.3$ (More consistent)

Set of male is having smaller value of coefficient of variation, male are more consistent or having less variability w.r.t. weight .

Difference between standard deviation and mean deviation:

- ✓ Standard deviation is calculated from mean only while mean deviation can be calculated from any average i.e. mean, median or mode.
- ✓ In mean deviation, algebraic signs are ignored, but in standard deviation, they are not.

Example 3.7: Sum and sum of squares corresponding to length X (in cm) and weight Y (in gm) of 50 fruit is $\sum X = 212$, $\sum X^2 = 910$, $\sum Y = 250$, $\sum Y^2 = 1260$. Which of the character (length or weight) is having more variability.

Solution:

Length

Mean $\overline{X} = \frac{212}{50} = 4.40$ cm

$\sigma_X^2 = \frac{910 - \frac{212^2}{50}}{50} = \frac{910 - 898.88}{50} = \frac{11.12}{50} = 0.22$ cm^2

$\sigma_x = 0.47$ cm

Coefficient of variation (X) = $\frac{0.47}{4.40}$ X 100 = 11.12%

Weight

Mean $\overline{Y} = \frac{250}{50} = 5.00$ gm

$\sigma_Y^2 = \frac{1260 - \frac{250^2}{50}}{50} = \frac{1260 - 1250}{50} = \frac{10.00}{50} = 0.20$gm^2

$\sigma_y = 0.45$ gm

Coefficient of variation (Y) = $\frac{0.45}{5.00}$ X 100 = 8.94%

Length is having more C.V.and is having more variability or less consistent whereas weight is having smaller variability or more consistent.

Example 3.8: Find mean and variance of first n natural number. What will be the mean and variance of first 20 natural numbers.

Solution:

First n natural numbers are 1,2,3,……..n

Sum of first n natural number = $\frac{n(n+1)}{2}$

Sum of square of first n natural number = $\frac{n(n+1)(2n+1)}{6}$

Mean = $\frac{n(n+1)}{2n} = \frac{n+1}{2}$

Variance = $\frac{1}{n}[\sum n^2 - \frac{(\sum n)^2}{n}]$

$$= \frac{1}{n}\left[\frac{n(n+1)(2n+1)}{6} - \frac{\frac{\{(n)(n+1)\}^2}{4}}{n}\right]$$

$$= \frac{n(n+1)}{n}\left[\frac{(2n+1)}{6} - \frac{n+1}{4}\right] = \frac{n(n+1)}{n}\frac{(4n+2-3n-3)}{12}$$

$$= \frac{(n+1)(n-1)}{12}$$

$$= \frac{n^2-1}{12}$$

Mean $= \frac{20+1}{2} = 10.5$

Variance $= \frac{400-1}{12} = 33.75$